Recreational and ergogenic substance use and substance use disorders in elite athletes: a narrative review

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ABSTRACT

Background Substances from various classes may be used for recreational purposes, self-treatment or to boost performance. When substance use shifts from occasional to regular, heavy or hazardous use, positive and negative effects can develop that vary by substance class and athlete. Regular use of recreational or performance enhancing substances can lead to misuse, sanctions or use disorders.

Objective To review the prevalence, patterns of use, risk factors, performance effects and types of intervention for all classes of recreational and performance enhancing substances in elite athletes by sport, ethnicity, country and gender.

Methods A comprehensive search was conducted to identify studies that compared the prevalence and patterns of substance use, misuse and use disorders in elite athletes with those of non-athletes and provided detailed demographic and sport variations in reasons for use, risk factors and performance effects for each main substance class.

Results Alcohol, cannabis, tobacco (nicotine) and prescribed opioids and stimulants are the most commonly used substances in elite athletes, but generally used at lower rates than in non-athletes. In contrast, use/misuse rates for binge alcohol, oral tobacco, non-prescription opioids and anabolic-androgenic steroids are higher among athletes than non-athletes, especially in power and collision sports. Cannabis/cannabinoids seem to have replaced nicotine as the second most commonly used substance.

Conclusions Substance use in elite athletes varies by country, ethnicity, gender, sport and competitive level. There are no studies on substance use disorder prevalence in elite male and female athletes and few studies with direct comparison groups.

INTRODUCTION

Substance use disorders are a major public health problem with high prevalence rates and associated significant economic, health and social burdens.[1] Elite athletes use and misuse recreational and performance enhancing substances (PES) for similar reasons as non-athletes.[2–14] Reasons for starting to use recreational substances include experimentation; socialisation; pleasure; self-treatment strategies, including stress relief, to reduce pain and negative emotions, increase alertness and energy and improve social self-confidence; and performance enhancement (ergogenic substance use).[2] PES are used to improve power, endurance, speed, aggression, agility, alertness and focus, and to reduce fatigue, injury healing time and percentage of body fat.[2 14] As use of recreational and/or PES increases and becomes problematic, impaired control, social problems, risky use, cravings and withdrawal symptoms may appear (figure 1).[2 15]

Substance use is typically defined as occasional social, recreational or episodic experimentation without associated problems.[1] Research studies commonly track trends in current use (past year or month) over time.[6] Substance misuse is defined as heavy, risky, harmful, hazardous, illicit, improper or problem use and denotes progression to a more regular and serious level of use.[1] Substance use disorders are defined as those which lead to clinically significant impairment or distress typically manifested by some of the following: excessive use and/or time spent using or recovering from use; unsuccessful efforts to cut down or quit; cravings and/or urges; failure to meet important obligations; continued use despite negative consequences; reduced time spent on important activities; physically hazardous use; and tolerance and/or withdrawal symptoms on abrupt cessation (figure 2).[16]

Alcohol, caffeine, nicotine, cannabis and stimulants are the most commonly used and misused substances by elite athletes and their non-athlete peers.[3 12 17–23] However, elite athletes have daily routines of intense physical and technical training in search of peak performance, which may lead to substance use that specifically targets performance, resulting in use of various ergogenic substances.[24–28] PES include anabolic agents such as anabolic-androgenic steroids (AAS) and selective androgen receptor modulators; peptide hormones, growth factors and related substances such as erythropoietin and growth hormone; β2 agonists; hormone and metabolic modulators, such as selective oestrogen receptor modulators; diuretics and masking agents; stimulants; and other substances that include opioids and cannabinoids.[14 26 27]

Furthermore, athletes may unwittingly ingest ergogenic substances by taking over-the-counter dietary supplements with ingredients prohibited by the World Anti-Doping Agency (WADA).[29] As defined by the International Olympic Committee, supplements are foods, food components, nutrients or non-food compounds ingested in addition to a habitually consumed diet to achieve a specific health and/or performance benefit[29] Dietary supplements come in many forms–sports foods, medical and ergogenic supplements, functional

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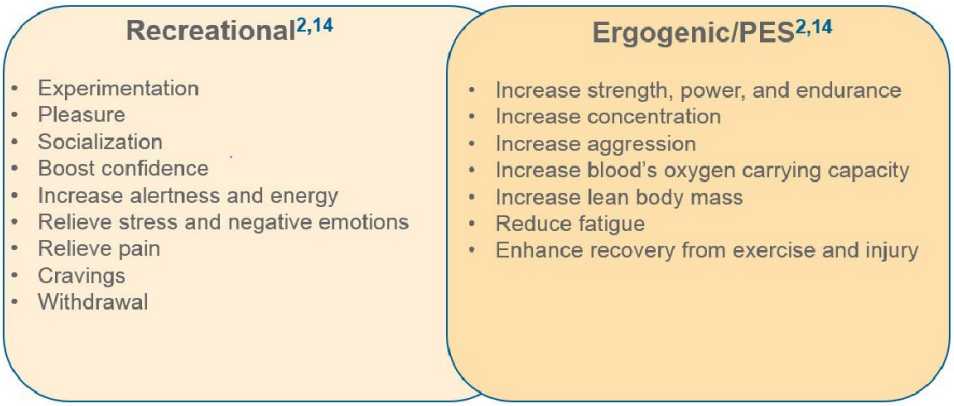

Figure 1 Reasons for recreational and ergogenic substance use/misuse. PES, performance enhancing substances.

foods and superfoods, herbal and botanical extracts and various forms of nutrients.[11 29 30] Active pharmaceutical agents in dietary supplements marketed for muscle building, weight loss and sexual enhancement may contain many banned products (eg, nandrolone, oxymetholone, oxandrolone, testosterone undecanoate, nandrolone decanoate and stanozolol).[25] The United States Food and Drug Administration identified 776 dietary supplements between 2007 and 2016 that contained unlabelled ingredients.[31] Selective androgen receptor modulators, which are banned by WADA,[32] may be included in some supplements but confusingly listed under various other names in the ingredients.[33] In addition to substances, athletes may use gene doping,[25 34] blood transfusions[24] and neuromodulation (eg, transcranial stimulation)[33 34] to boost performance. Gene doping is the introduction of genetic or other materials into the cells of athletes to increase the natural biosynthesis of performance enhancing proteins such as erythropoietin or insulin-like growth factor 1.

This review aims to summarise the literature on prevalence, risk factors, performance effects, assessment, and treatment of substance use, misuse and use disorders in elite athletes. In addition, we seek to identify and discuss studies that report: (1) prevalence rates in men and women with direct or indirect comparison groups; (2) differences in use, misuse and use disorders by sport, level of competition, country, ethnicity and gender; (3) positive and negative effects of recreational and performance enhancing substances; and (4) risk factors and preventive intervention strategies.

METHODS

The literature was searched systematically by an experienced health systems librarian (MEH). The databases PubMed, Scopus, SPORTDiscus, PsychINFO and Cochrane were searched using the following key words (MeSH search terms were used for general topics but not for individual substance classes): "substance-related disorders"(MeSH) OR "substance use"(tw) OR "recreational drugs" OR marijuana OR methamphetamine OR "substance abuse" OR "prescription drug" OR "intravenous drug" OR steroids OR heroin (tw) OR "performance enhancing" OR non-medical OR cocaine OR amphetamine OR "illegal drugs" OR "drug addiction" OR addiction (tw)) AND (("Athletes"(MeSH) OR athlete OR athletic OR athletes) AND (elite OR professional OR Olympic OR "highly trained" OR "high performing" OR "high performance" OR competitive OR collegiate))). An initial search was conducted on 1 March 2018 for relevant articles in English from 2000 and later and a follow-up search was conducted on 22 October 2018 looking for articles published after 1 March 2018. Four experienced addiction and sports psychiatrists (DM, TS, JMC-M, CLR) and one sports neurologist (BH) with expertise in substance use/use disorders reviewed both lists and then abstracts of studies matching the inclusion criteria (see below). We defined elite athletes as those competing at professional, Olympic, or collegiate/university levels.

Articles cited in this manuscript include: (1) prevalence studies of past year and current substance use with direct and

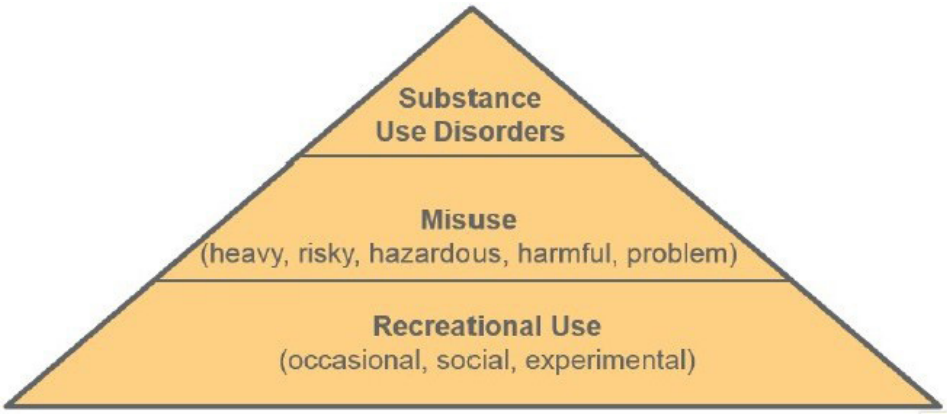

Figure 2 Continuum of substance use.

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indirect comparison groups; (2) prevalence studies of risky drinking and adverse alcohol behaviours; (3) studies, reviews, and meta-analyses of positive and negative recreational and ergogenic substance effects on sports performance; (4) models of screening, diagnosis, and treatment of substance misuse and use disorders; and (5) prevention and intervention studies with binge/risky drinkers, oral tobacco users, and illicit drug users. Some books or book chapters were included when they represented important expert consensus opinion of otherwise unstudied areas. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) approach was not utilised because of the breadth of this paper and the absence of high quality studies across substance classes, spectra of use, sports, and countries. Only the most commonly used and misused substances will be covered.

RESULTS

The first search (1 March 2018) yielded a total of 1655 titles, and the second search (22 October 2018) yielded an additional 93 titles for a total of 1748. Out of this list, 130 articles that met our inclusion criteria were selected for further review. Ninety were determined to be relevant for this paper, and an additional 47 were included after a review of the references from the original 90 and a review of the non-sporting literature for essential references on general substance topics.

General prevalence of substance use, misuse and use disorders

In most sports, elite athletes use substances at lower rates than the general population, especially during the competitive season.[2 3 6 8 13 17 18 35–38] Specifically, collegiate athletes across all United States sports report lower annual use of alcohol, cigarettes, marijuana, amphetamines, AAS, cocaine, ecstasy, and lysergic acid diethylamide than their non-athlete peers.[6 35 36] Yet some substances (eg, alcohol, cannabis, nicotine, and prescribed stimulants/opioid pain medications, with or without a prescription) are used/misused at higher rates by male and female athletes in certain sports.[6 8–10 17 38–53] Specifically, US male athletes participating in lacrosse, ice hockey, rugby, wrestling, and swimming at elite levels report annual binge drinking rates of 50% to 70%, while US female athletes in lacrosse, ice hockey, and swimming at elite levels report rates of 49% to 57%.[6 52] Binge drinking, especially associated with adverse intoxication behaviours, is even higher for white men, non-heterosexuals, problem gamblers, and/or PES-using athletes.[3 42–44 54–61] Professional European male football (soccer) players from five countries (N=540)[47] and professional male rugby players from eight international countries from four continents and the Pacific Islands (N=990) reported cross-sectional rates of recent adverse alcohol behaviours (regular, heavy drinking, and/or binge drinking) using the AUDIT-C ranging from 6% to 17% and 8% to 21%, respectively.[62] A study of elite male Rugby League players from Australasia, also using the AUDIT-C showed much higher rates of recent hazardous alcohol use than in the general population both before (68.6%; N=404) and during (62.8%; N=278) the season.[52] A later longitudinal study by Zanotti, *et al* using the Psychiatric Diagnostic Survey Questionnaire (N=270) showed combined presumptive current rates of Diagnostic and Statistical Manual of Mental Disorders IV alcohol abuse/dependence of 7.2% to 10.3%[62] among US male and female college athletes.[63]

Although self-report surveys and competition-day urine drug testing are the most common ways of determining use or misuse, they probably underestimate prevalence.[8 63 64] More reliable measures include athlete biological passport baselines (obtaining an individual's profile of biological markers of doping over time); team urine surveillance and post-game testing; repeat testing; hair testing; early out-of-season testing; doping attitude scale administration (assesses risk perspectives on use, with riskier views associated with increased likelihood of use); indirect questioning techniques (eg, "Would you consider using cannabis or AAS if the health risk were low or there was no urine testing?"); and interviews with athletes, teammates, parents and/or coaches.[63–68]

Cannabis

Cannabis containing primarily tetrahydrocannabinol (THC) is the most widely used illicit substance in the general population and in elite athletes,[5 6] although elite athletes overall use cannabis less than the general population or non-athlete peers.[6 51 69] Self-reported annual rates of cannabis use have risen from about 20% to 25% among US college athletes over the past decade, with higher rates among division III versus division I athletes (33% vs 18%) and those living in states where medical or recreational cannabis is legal vs illegal (39% vs 26%).[6]

There is also a trend in other countries for increased cannabis use in elite athletes such that it has replaced nicotine as the second most widely used substance.[5 17 39] No specific data are available on cannabis use rates in professional or Olympic athletes outside the United States. More recently, however, cannabis products containing cannabidiol have become available via medical dispensaries and the internet and are advertised for pain control, insomnia, stress anxiety, anger and depression. Many of these products (37%) state inaccurate amounts of cannabidiol, and 21.43% contained low but detectable levels of THC that could result in a positive test and/or sanctions.[70]

Synthetic cannabis

Synthetic cannabinoids, which have emerged over the past decade, have also been detected with increased frequency in elite athletes,[71 72] including US collegiate athletes.[6] Year-round trends for athletes worldwide are uncertain. Synthetic cannabinoids may often be neurotoxic and appear to share psychoactive properties with THC.[73] Athletes who are male, current drinkers and who have used hookah and marijuana in the past month are more likely to use synthetic cannabinoids. However, estimated synthetic cannabinoid use by US collegiate athletes of their team mates is reportedly overestimated, and this overestimation is associated with increased personal use of synthetic cannabinoids.[74]

Cocaine

Despite a low prevalence of cocaine use in elite athletes (0.1–3.8%), it is the second most prevalent illicit drug used.[6 8 75] However, in some US collegiate sports (eg, lacrosse, ice hockey, swimming and wrestling), annual rates of use are significantly higher (7–22%).[6 8 75] Relatively more uncommon or unknown stimulants have been detected in several supplements, reportedly leading to disqualification for substances such as N, α-diethyl-phenylethylamine, a methamphetamine analogue.[76] Athletes seem to be more likely than non-athletes to consume these disguised stimulants.[76 77] Other stimulants such as synthetic cathinones (chemically similar to ephedrine), sold as "bath salts" in the USA, were reportedly used more often by university athletes than their non-athlete peers.[77]

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Nicotine

Nicotine, whether smoked (eg, as cigars, cigarettes or hookahs), used orally (eg, via moist snuff, snus or leaf tobacco) or vaped, is also widely used by elite athletes, especially US male lacrosse, baseball, and ice hockey players.[6] Rates of annual use range from 44% to 46%, with 20% reporting daily use.[6] For female US collegiate athletes, the highest rate of use was 13% for ice hockey players.[6] In addition, 8% of US collegiate student athletes reported use of nicotine through vaping devices in the past year.[6] The only current data on rate of use by professional/Olympic athletes comes from US baseball players (35% for snuff), Finnish Olympic athletes (25% for snuff), current and former professional football (soccer) players worldwide (7% and 12% smoking, respectively) and recently retired professional rugby players (15% for smoking).[19 78–80]

Prescription drugs

Prescription medication misuse occurs most often with stimulants prescribed for attention-deficit/hyperactivity disorder or opioids prescribed for acute or chronic pain.[2 10 81] In 2017, US collegiate athletes reported past-year stimulant medication use of 7.5% without a prescription (6.7% in 2009), and 6.6% with a prescription.[6] Also in 2017, 2.9% of US collegiate athletes reported past-year use of opioids without a prescription, down from 5.1% in 2009. Non-prescription opioid past-year rates of use are even higher for male (17.9%) and female (10.6%) US collegiate athletes who are injured, than for those who are neither athletes nor injured (compared with 7.9% for men and 6.7% for women in the general population).[10]

Performance enhancing substances (AAS)

The lifetime or current prevalence of PES use (ie, doping) varies greatly (1–70%) depending on the sport, substance, detection method and definition used.[7 82] Self-report surveys are the most common method of determining use, and the accuracy of these, especially with AAS, has been questioned by several authors.[7 82 83] Each year, 1–2% of WADA urine and blood tests are positive for PES, with 48% of these violations being for AAS by elite male and female athletes.[7] A 2013 and 2017 National Collegiate Athletic Association (NCAA) self-report study of approximately 20 000 US collegiate athletes showed that 0.4% of all respondents from both studies across all sports and divisions self-reported past-year use of AAS.[6 82] Since these annual use rates are surprisingly lower than in the general population, it raises questions about the accuracy of self-reported prevalence.

Kersey *et al*[83] reported higher AAS lifetime prevalence rates in elite athletes who participate in power and collision sports compared with other sports, including a 1989 study (N=176),[84] which showed that 37.5% of male and female competitive body builders and 66.7% of elite male power lifters (N=15), respectively, admitted AAS use.[85] In addition, a 2009 study of retired National Football League players (N=2552)[86] reported that 9.1% admitted using AAS during their professional careers, with the highest rates among offensive (16.3%) and defensive (14.8%) linemen. Finally, a 30-year study of 996 retired male Swedish power sport athletes disclosed a lifetime prevalence of AAS use of 21%.[49] An extensive review of doping rates by de Hon and colleagues[7] argues that data gathered from questionnaires using the randomised response technique coupled with biological parameters provide more accurate estimates of doping prevalence, with results ranging from 14% to 39% among elite athletes, with notable variation between subgroups of sport.[7]

Between 40% and 100% of all athletes may use supplements, depending on the type of sport, level of competition and the definition of supplements.[11] Considering only elite athletes, 10 274 college and other elite athletes were included in a meta-analysis that found that 46% of college athletes and 59% of other elite athletes used supplements.[87] Among 310 competitors at the International Association of Athletics Federations World Championships (in track and field), 86% reported supplement use (71% to improve recovery; 52% to improve health; 46% to improve performance).[88] These high rates of supplement use by elite athletes is consistent across other studies.[89–92] Therefore, many elite athletes run some risk of testing positive for a prohibited substance since banned substances may be present as accidental or known contaminants.

Disorders associated with substance use

We found no studies in elite athletes that systematically analysed the prevalence of current substance use disorder using structured clinical interviews. However, expert opinion, based largely on literature review, suggests that rates for moderate or severe use disorders across substances are relatively low in active athletes and possibly relatively more common in retired athletes.[4 16 47 93–96] Mild to moderate nicotine and alcohol use disorders in older elite athletes may be relatively common as identified through comprehensive preseason screening and treatment.[2 16 93 97] Finally, based on 10 international studies, about 30% of AAS users develop a steroid use disorder that appears to be caused and/or perpetuated by three main factors, specifically: (1) a strong desire to maintain a certain physique; (2) suppression of the hypothalamic-pituitary-gonadal axis, causing testosterone production to shut down; and (3) attempts to reverse common side effects of use (ie, loss of libido, impaired erectile function, decreased energy and depression).[28]

Factors associated with substance use and misuse

Common longitudinal factors associated with substance use and misuse include: (1) sport context and culture (eg, normative beliefs about heavy peer drinking or illicit drug use); (2) situational temptation (eg, availability of alcohol or other drugs); (3) permissive attitudes among athletes, coaches and parents; (4) sensation-seeking personality; and (5) male sex.[98–101]

Cross-sectional factors associated with use and misuse include: (1) current use of performance enhancing substances or tobacco; (2) identification as lesbian, gay, bisexual or transgender; (3) party lifestyle or drinking game participation; (4) overestimating peer use; (5) achievement orientation; (6) lower use of protective measures (eg, avoiding serious intoxication, using a designated driver); (7) leadership position; (8) fraternity/sorority membership[43–46 57 98–100 102]; (9) problem gambling[60]; and (10) injury.[10] Strong religious beliefs have been inversely associated with use of alcohol, marijuana and other drugs.[101]

Substance effects on performance

Alcohol

Some athletes may use alcohol before a competition to reduce anxiety or tremor and improve subjective self-confidence.[45 100 103] Alcohol is no longer a prohibited substance for competition in specified sports (ie, air sports, automobile, archery and power boating).[32] However, many US professional teams have banned alcohol on flights after away games and in the team facility to reduce the risk of drinking and driving. After a competition, alcohol is used by some athletes to reduce stress, boost self-esteem, increase social connectedness, improve

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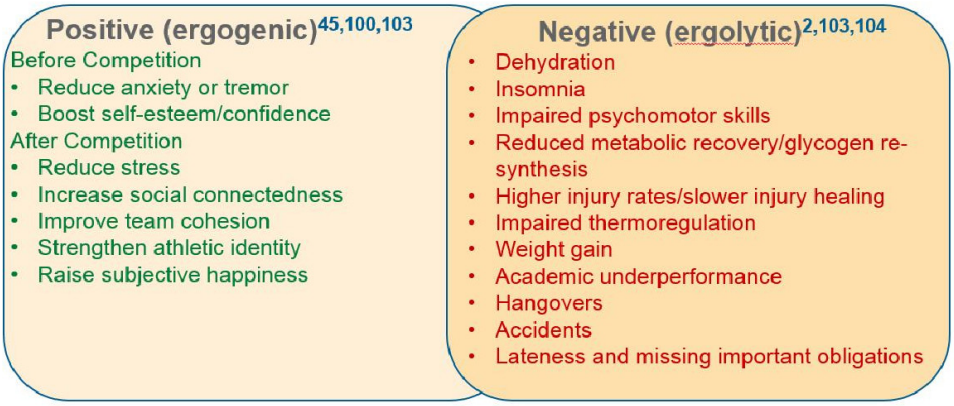

Figure 3 Alcohol: positive (ergogenic) and negative (ergolytic) effects.

team cohesion, strengthen athletic identity and raise subjective happiness, especially in sports with strong cultural connections to it—for example, lacrosse, rugby, football (soccer) and US football.[45 100 103] Ergolytic effects of alcohol include dehydration, insomnia, higher injury rates, slower injury healing, impaired psychomotor skills, hangovers, accidents, lateness, missing important obligations, reduced metabolic recovery/glycogen resynthesis, impaired thermoregulation, weight gain and academic underperformance that can threaten athletic eligibility (figure 3).[2 103 104]

Cannabis

Cannabis contains delta-9-THC, cannabidiol, and several other cannabinoids, generating different effects in different athletes.[5 93] Athletes report positive ergogenic effects from cannabis use, such as improved sleep and reduced levels of pain and anxiety,[104] but this is not generally supported by scientific evidence. Several potentially ergolytic effects include increased heart rate and blood pressure; slower reaction time; impaired coordination; reduced motivation; and increased likelihood of anxiety, psychosis and delirium.[69 72 104] Other perceived positive reasons for its use include socialisation and pleasure.[41 69] Additionally, athletes self-report use of cannabis for pain and management of concussion.[105]

Stimulants

Elite athletes who take stimulants (ie, caffeine, nicotine and prescription stimulants such as methylphenidate or amphetamine) may do so to improve reaction time and concentration, increase arousal, improve memory, boost energy, trigger relaxation and confidence and improve energy when fatigued.[2 19 104 106–108] The negative effects of stimulants, however, are more apparent in high doses or when combined ("stacked"), both of which may be relatively common among male athletes who use performance enhancing substances.[2 42 43] Athletes may knowingly or unknowingly consume large amounts of caffeine in dietary supplements; larger doses do not appear to increase performance and are more likely to cause side effects.[29] Ergolytic effects of high-dose stimulant use or combination stacking may include anxiety, insomnia, gastric irritation, tachycardia and tremors (figure 4).[2 21 104 106]

Nicotine

Nicotine is currently being studied by WADA—in part because of its potential ergogenic effects on performance.[32] Athletes' perceptions of nicotine's ergogenic effects include improvements in alertness and concentration, energy and focus, muscular strength and power, endurance, relaxation/calmness and weight control, as well as reduction of boredom. Studies on the effects of nicotine suggest possible ergogenesis via increased blood flow to muscles, improved endurance, lipolysis and improved cognition and reaction time.[18 105] Ergolytic effects may include elevated blood pressure, anxiety, insomnia and chronic respiratory infections (figure 5).[2 19 104]

Anabolic-androgenic steroids (AAS)

Ergogenic effects of AAS include enhanced muscle mass, improved biomechanical efficiency and anticatabolic effects.[109] Potential ergolytic effects include cognitive impairment,[110]

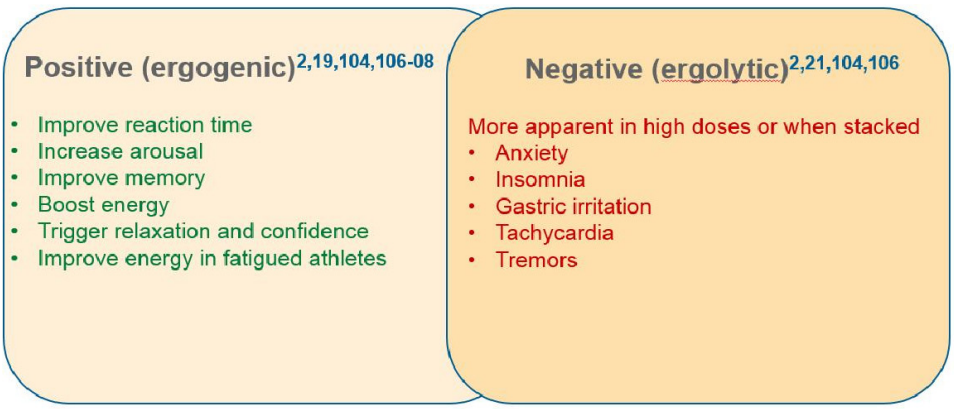

Figure 4 Stimulants: ergogenic and ergolytic effects.

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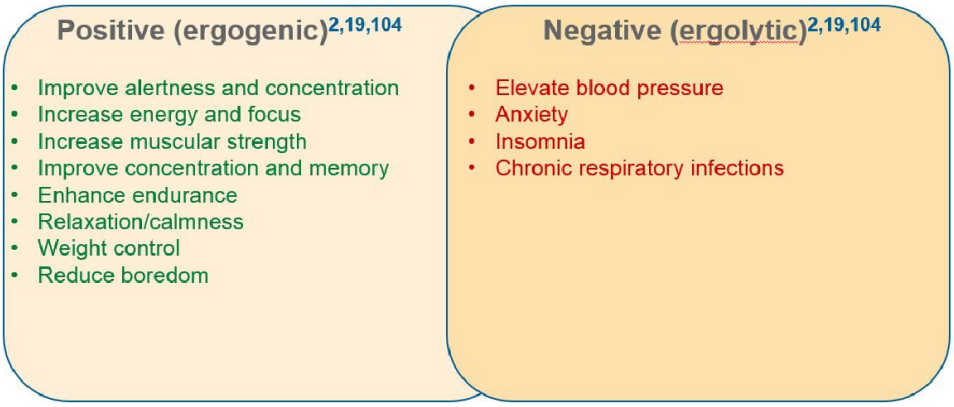

Figure 5 Nicotine: ergogenic and ergolytic effects.

negative mood symptoms,[14] psychosis,[14] aggression[14] and injury, especially tendon rupture.[111] The most commonly reported adverse effects of AAS in female athletes include hirsutism, alopecia, deepening of the voice, clitoromegaly, menstrual disturbance and aggression.[112] In male athletes, the most common side effects are acne, gynaecomastia, testicular atrophy, libido and fertility changes, impotence, behavioural problems, lipid changes and stretch marks.[25 113] Long-term side effects may be severe and depend on the dosage and duration. Most noteworthy are death from cardiovascular disease, sterility, masculinisation, addiction, liver problems, immune system changes and psychiatric manifestations.[25 114] Cycles of using anabolic-androgenic steroids are usually interspersed with either drug-free periods and/or periods that are sometimes associated with other drug use (eg, diuretics, benzodiazepines, growth hormone, insulin, or insulin-like growth factor 1) to mitigate side effects[2 109] or optimise muscle growth, recovery time,and performance.[27] These other drugs may also have ergolytic effects.[115–118] Finally, some individuals who use AAS have muscle dysmorphia, a preoccupation that one's body is not sufficiently lean and muscular, often with significant body image distortion.[119–121]

Prevention and treatment

Elite athletes may undergo formal urine and blood testing by national or international antidoping organisations.[6 32] However, detection of prohibited PES and performance enhancing methods remains a challenge. Elite athletes may use many strategies to avoid detection, including use of unknown substances or substances with unknown structures/metabolites ("designer steroids"), enzyme inhibitors, masking agents to lower other drug concentrations and naturally occurring substance substitution (eg, synthetic testosterone).[68]

To deal with some of the shortcomings of national or international drug testing, the athlete biological passport is starting to be used more widely, and consists of an individual electronic record for athletes in which profiles of biological markers of doping and results of doping tests are collated over a period of time; doping violations are detected by noting variances from an athlete's established levels outside permissible limits.[25] In addition, some elite athletes are also closely monitored for early signs of problematic substance use by coaches, team physicians and trainers.[93]

Interventions

Interventions for regular, risky or disordered use of substances in elite athletes have not been well studied. Nonetheless, approaches to reducing spit tobacco use among baseball players,[78 122 123] and newer approaches to service delivery models for mental health and substance use disorders in elite athletes, provide some guidance.[97 124] For spit tobacco use, two approaches used preseason physical and dental examinations as an opportunity to screen for heavy nicotine use and examine players for oral lesions.[122 123] Screening included the administration of the nicotine dependence rating scale (Fagerstrom) and an oral dental examination. If either screen was positive, then a brief intervention by a dental technician or a substance counsellor experienced in tobacco cessation was delivered. In one study,[78] after 10 years of annual oral examinations and dental technician interventions, spit tobacco use dropped from 41.1% to 25.6% of players.

In more recent models of service delivery, experienced sports clinicians are onsite with teams, to integrate substance screenings and brief interventions with other health screenings and interventions.[97 124] One model uses experienced mental health/substance providers, who work with the team all year, to conduct screenings for substance use/misuse (and symptoms of other common mental health disorders) at the preseason physical. They then carry out follow-up evaluations and treatments for those who screen positive. This approach has led to increasing service utilisation rates and problem identification at earlier stages, when problems are easier to treat.[2 97]

Finally, with drug testing a known deterrent to substance use among athletes,[125] some research suggests that increased frequency of drug testing, including during high-risk time periods (eg, immediately after a game when athletes are socialising or shortly after a season ends), might reduce illicit drug use among elite athletes.[65]

Brief individual or group interventions using motivational interviewing strategies delivered by clinicians, sometimes with key family members or members of the athlete's entourage in attendance, may successfully prevent or diminish binge drinking or other substance use in college athletes.[126–129] Similarly, binge drinking may be successfully reduced when athletic trainers and academic advisers screen for and deliver brief motivational interventions.[130] Group therapy, as an adjunct to medication treatment, is effective when dealing with substance use disorders among professional athletes.[5 52 131] No research into the use of self-help groups such as Alcoholics Anonymous or Narcotics Anonymous or intensive outpatient or residential treatment of elite athletes has been carried out. Although these treatments are evidence-based in other populations, concerns about confidentiality may limit elite athletes' willingness to participate in self-help groups.[4]

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Medications

No pharmacological studies on treatment of substance use disorders in elite athletes have been carried out. Such treatment is generally grounded in psychosocial modalities, and pharmacotherapy is used to manage withdrawal and cravings and to treat comorbid mental health symptoms or disorders such as insomnia, anxiety or depression.[4] Since alcohol is the most commonly misused substance among elite athletes, the use of naltrexone, acamprosate or disulfiram should be considered when cravings are strong or persistent or when the athlete cannot stop drinking.[132] There are unique implications for treatment of opioid use disorder in elite athletes because the treatment often involves use of opioid agonists (eg, methadone and buprenorphine). Since opioids are prohibited *in-competition* by WADA,[32] elite athletes falling under WADA's governance require a break from training and competition if agonist therapy is used.[132] Thus, providers might consider the use of an opioid antagonist (oral or extended-release naltrexone) for those with an opioid use disorder[133] when a break in training or competition is not possible. Since heavy oral nicotine use is seen frequently in some sports and can be accompanied by tolerance, physiological dependence and cravings, pharmacological strategies for moderate to severe nicotine use disorders in elite athletes should be considered. These include nicotine replacement therapy with or without bupropion, varenicline alone or varenicline plus bupropion.[134 135] Electronic cigarettes or nicotine vaping devices are not recommended for use in elite athletes, since their safety and effects on respiratory function have not been established.[136 137]

What is already known?

- Self-report and competition-day urine testing probably underestimate substance use and misuse among elite athletes.
- Substance use among elite athletes varies considerably by gender, sexual orientation, ethnicity, country, sport and age/level of competition and is generally lower during the playing season than the off-season.
- Binge drinking and adverse drinking-related behaviours are more common in US collegiate athletes (especially white men in collision and/or contact sports) than in non-athletes.
- Nicotine use via oral tobacco is higher in elite athletes in some sports—for example, baseball, ice hockey and lacrosse, than in non-athletes.
- Stimulants, anabolic-androgenic steroids and other body mass builders are the most common substances identified in competition testing.

What are the new findings?

- Binge drinking rates are higher in elite athletes at the professional level (compared with community populations) and increase further with transition into retirement.
- Cannabis/cannabinoids may have replaced nicotine as the second most commonly misused substance in collegiate and some professional athletes, especially in areas where they are legal.
- Non-medical use of prescription opioids should be suspected in injured male athletes, especially in the USA.
- Newer substance use/misuse identification strategies such as the biological passport, hair testing, attitude scale administration, early out-of-season testing, total team testing, postgame testing and direct/indirect questioning of athletes and their teammates, parents and coaches are likely to more accurately reflect current or past use and should be adopted.

CONCLUSIONS

Prevalence studies (especially with women) with non-athlete comparison groups are needed to clarify relative rates of use and misuse among elite athletes. Studies should determine the prevalence of current and past substance use disorders using structured clinical interviews. More research is also needed to ascertain short- and long-term consequences of alcohol, cannabis, PES, nicotine, and prescription stimulants/opioids by elite athletes. Screening instruments for cannabis, other illicit drugs, prescription stimulants and opioids, and PES need to be validated within athlete populations. Once instruments are validated, preseason screening should be conducted routinely, and the outcomes of interventions with athletes who test positive should be studied. Research is also needed to delineate risk factors for use and misuse at all competitive levels, across genders and countries, so that new preventive interventions can be designed, implemented and studied. We recommend that more recently described approaches for identifying substance use (eg, onsite mental health/substance care, more frequent urine testing, postgame testing, hair testing and biological passport profiling) should be implemented on a larger scale, with accompanying effectiveness studies.

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REFERENCES

1 United States Department of Health and Human Services Office of the Surgeon General. Facing addiction in America: The Surgeon General's report on alcohol, drugs and health. In: *United States Department of Health and Human services OotSG*. Washington DC: United States Department of Health and Human Services, Office of the Surgeon General, 2016.
2 McDuff DR, Baron D. Substance use in athletics: a sports psychiatry perspective. *Clin Sports Med* 2005;24:885–97.
3 Yusko DA, Buckman JF, White HR, *et al*. Risk for excessive alcohol use and drinking-related problems in college student athletes. *Addict Behav* 2008;33:1546–56.
4 Morse ED. Substance use in athletes. In: Baron DA, Cl R, Baron SH, eds. *Clinical sports psychiatry: an international perspective*. Oxford, UK: Wiley, 2013.
5 Brisola-Santos MB, Gallinaro JGdeMe, Gil F, *et al*. Prevalence and correlates of cannabis use among athletes: a systematic review. *Am J Addict* 2016;25:518–28.

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6 NCAA. National study on substance abuse habits of college student-athletes, 2018. Available: http://www.ncaa.org/sites/default/files/2018RES_Substance_Use_Final_Report_FINAL_20180611.pdf [Accessed November 17 2018].
7 de Hon O, Kuipers H, van Bottenburg M. Prevalence of doping use in elite sports: a review of numbers and methods. *Sports Med* 2015;45:57–69.
8 Dunn M, Thomas JO, Swift W, *et al*. Elite athletes' estimates of the prevalence of illicit drug use: evidence for the false consensus effect. *Drug Alcohol Rev* 2012;31:27–32.
9 Ford JA. Substance use among college athletes: a comparison based on sport/team affiliation. *J Am Coll Health* 2007;55:367–73.
10 Ford JA, Pomykacz C, Veliz P, *et al*. Sports involvement, injury history, and non-medical use of prescription opioids among college students: an analysis with a national sample. *Am J Addict* 2018;27:15–22.
11 Garthe I, Maughan RJ. Athletes and supplements: prevalence and perspectives. *Int J Sport Nutr Exerc Metab* 2018;28:126–38.
12 Green GA, Uryasz FD, Petr TA, *et al*. NCAA study of substance use and abuse habits of college student-athletes. *Clin J Sport Med* 2001;11:51–6.
13 Lisha NE, Sussman S. Relationship of high school and college sports participation with alcohol, tobacco, and illicit drug use: a review. *Addict Behav* 2010;35:399–407.
14 Reardon C, Creado S. Drug abuse in athletes. *Subst Abuse Rehabil* 2014;5:95–105.
15 American Psychiatric Association. What is addiction? 2017. Available: https://www.psychiatry.org/patients-families/addiction/what-is-addiction [Accessed 2 April 2019].
16 American Psychiatric Association. *Diagnostic and statistical manual of mental disorders (DSM-5)*. Washington, DC: American Psychiatric Publishing, 2013.
17 Diehl K, Thiel A, Zipfel S, *et al*. Substance use among elite adolescent athletes: findings from the goal study. *Scand J Med Sci Sports* 2014;24:250–8.
18 Momaya A, Fawal M, Estes R. Performance-enhancing substances in sports: a review of the literature. *Sports Med* 2015;45:517–31.
19 Mündel T. Nicotine: sporting friend or foe? A review of athlete use, performance consequences and other considerations. *Sports Med* 2017;47:2497–506.
20 Marclay F, Grata E, Perrenoud L, *et al*. A one-year monitoring of nicotine use in sport: frontier between potential performance enhancement and addiction issues. *Forensic Sci Int* 2011;213:73–84.
21 Cappelletti S, Piacentino D, Daria P, Sani G, *et al*. Caffeine: cognitive and physical performance enhancer or psychoactive drug? *Curr Neuropharmacol* 2015;13:71–88.
22 Drobnic F, Galilea PA. Doping control adverse results prevalence worldwide for 13 consecutive years. *Apunts. Medicina de l'Esport* 2018;53:11–18.
23 Kabiri S, Cochran JK, Stewart BJ, *et al*. Doping among professional athletes in Iran: a test of Akers's social learning theory. *Int J Offender Ther Comp Criminol* 2018;62:1384–410.
24 Hughes D. The world anti-doping code in sport: update for 2015. *Australian Prescriber* 2015;38:167–70.
25 Mazzeo F, Monda V, Santamaria S, *et al*. Antidoping program: an important factor in the promotion and protection of the integrity of sport and athlete's health. *J Sports Med Phys Fitness* 2018;58.
26 Siebert DM, Rao AL. The use and abuse of human growth hormone in sports. *Sports Health* 2018;10:419–26.
27 Anderson LJ, Tamayose JM, Garcia JM. Use of growth hormone, IGF-I, and insulin for anabolic purpose: pharmacological basis, methods of detection, and adverse effects. *Mol Cell Endocrinol* 2018;464:65–74.
28 Kanayama G, Pope HG. History and epidemiology of anabolic androgens in athletes and non-athletes. *Mol Cell Endocrinol* 2018;464:4–13.
29 Maughan RJ, Burke LM, Dvorak J, *et al*. IOC consensus statement: dietary supplements and the high-performance athlete. *Br J Sports Med* 2018;52:439–55.
30 Hoyte CO, Albert D, Heard KJ. The use of energy drinks, dietary supplements, and prescription medications by United States college students to enhance athletic performance. *J Community Health* 2013;38:575–80.
31 Tucker J, Fischer T, Upjohn L, *et al*. Unapproved pharmaceutical ingredients included in dietary supplements associated with US Food and Drug Administration warnings. *JAMA Network Open* 2018;1:e183337.
32 World Anti-Doping Agency (WADA). World anti-doping agency prohibited list, 2019. Available: https://www.usada.org/substances/prohibited-list/
33 Geyer H, Schänzer W, Thevis M. Anabolic agents: recent strategies for their detection and protection from inadvertent doping. *Br J Sports Med* 2014;48:820–6.
34 van der Gronde T, de Hon O, Haisma HJ, *et al*. Gene doping: an overview and current implications for athletes. *Br J Sports Med* 2013;47:670–8.
35 Schulenberg JE, Johnston LD, O'Malley PM, *et al*. Monitoring the future national survey results on drug use, 1975-2016. Volume II, College students & adults ages 19-55: Institute for Social Research. *University of Michigan* 2017.
36 SIUC/Core Institute. Annual Data. Core Alcohol and Drug Survey Long Form - Form 194. *Executive Summary. Carbondale, IL: SIUC/Core Institute* 2013;2013 https://core.siu.edu/_common/documents/2013.pdf
37 Heather N, Partington S, Partington E, *et al*. Alcohol use disorders and hazardous drinking among undergraduates at English universities. *Alcohol Alcohol* 2011;46:270–7.
38 Lisha NE, Crano WD, Delucchi KL. Participation in team sports and alcohol and marijuana use initiation trajectories. *J Drug Issues* 2014;44:83–93.
39 Kondric M, Sekulic D, Mandic GF. Substance use and misuse among Slovenian table tennis players. *Subst Use Misuse* 2010;45:543–53.
40 Sagoe D, Molde H, Andreassen CS, *et al*. The global epidemiology of anabolic-androgenic steroid use: a meta-analysis and meta-regression analysis. *Ann Epidemiol* 2014;24:383–98.
41 Buckman JF, Yusko DA, Farris SG, *et al*. Risk of marijuana use in male and female college student athletes and nonathletes*. *J. Stud. Alcohol Drugs* 2011;72:586–91.
42 Buckman JF, Yusko DA, White HR, *et al*. Risk profile of male college athletes who use performance-enhancing substances. *J Stud Alcohol Drugs* 2009;70:919–23.
43 Buckman JF, Farris SG, Yusko DA. A national study of substance use behaviors among NCAA male athletes who use banned performance enhancing substances. *Drug Alcohol Depend* 2013;131:50–5.
44 Green K, Nelson TF, Hartmann D. Binge drinking and sports participation in college: patterns among athletes and former athletes. *International Review for the Sociology of Sport* 2014;49:417–34.
45 Zhou J, Heim D, O'Brien K. Alcohol consumption, athlete identity, and happiness among student sportspeople as a function of sport-type. *Alcohol Alcohol* 2015;50:617–23.
46 Veliz PT, Boyd CJ, Epstein-Ngo Q, *et al*. Substance use among sexual minority collegiate athletes: a national study. *Drug Alcohol Depend* 2015;156:e230.
47 Gouttebarge V, Backx FJ, Aoki H, *et al*. Symptoms of common mental disorders in professional football (soccer) across five European countries. *J Sports Sci Med* 2015;14.
48 Al Ghobain M, Konbaz MS, Almassad A, *et al*. Prevalence, knowledge and attitude of prohibited substances use (doping) among Saudi sport players. *Subst Abuse Treat Prev Policy* 2016;11.
49 Lindqvist AS, Moberg T, Eriksson BO, *et al*. A retrospective 30-year follow-up study of former Swedish-elite male athletes in power sports with a past anabolic androgenic steroids use: a focus on mental health. *Br J Sports Med* 2013;47:965–9.
50 Pereira HMG, Sardela VF. Stimulant doping agents used in Brazil: prevalence, detectability, analytical implications, and challenges. *Subst Use Misuse* 2014;49:1098–114.
51 Peretti-Watel P, Guagliardo V, Verger P, *et al*. Sporting activity and drug use: alcohol, cigarette and cannabis use among elite student athletes. *Addiction* 2003;98:1249–56.
52 Du Preez EJ, Graham KS, Gan TY, *et al*. Depression, anxiety, and alcohol use in elite rugby league players over a competitive season. *Clin J Sport Med* 2017;27:530–5.
53 Rigotti NA, Lee JE, Wechsler H. US college students' use of tobacco products: results of a national survey. *JAMA* 2000;284.
54 Leichliter JS, Meilman PW, Presley CA, *et al*. Alcohol use and related consequences among students with varying levels of involvement in college athletics. *J Am Coll Health* 1998;46:257–62.
55 Nelson TF, Wechsler H. Alcohol and college athletes. *Med Sci Sports Exerc* 2001:43–7.
56 Wilson GS, Pritchard ME, Schaffer J. Athletic status and drinking behavior in college students: the influence of gender and coping styles. *J Am Coll Health* 2004;52:269–75.
57 Grossbard J, Geisner IM, Neighbors C, *et al*. Are drinking games sports? College athlete participation in drinking games and alcohol-related problems. *J. Stud. Alcohol Drugs* 2007;68:97–105.
58 Wechsler H, Davenport AE, Dowdall GW, *et al*. Binge drinking, tobacco, and illicit drug use and involvement in college athletics. *J Am Coll Health* 1997;45:195–200.
59 Barry AE, Howell SM, Riplinger A, *et al*. Alcohol use among college athletes: do intercollegiate, Club, or intramural student athletes drink differently? *Subst Use Misuse* 2015;50:302–7.
60 Huang J-H, Jacobs DF, Derevensky JL, *et al*. Gambling and health risk behaviors among U.S. college student-athletes: findings from a national study. *J Adolesc Health* 2007;40:390–7.
61 O'Brien KS, Ali A, Cotter JD, *et al*. Hazardous drinking in New Zealand sportspeople: level of sporting participation and drinking motives. *Alcohol Alcohol* 2007;42:376–82.
62 Zanotti D, Kaier E, Strunk K. Prevalence of mental health symptoms in college athletes. In: . Washington, D.C: American Psychological Association, 2017.
63 Druckman JN, Gilli M, Klar S, *et al*. Measuring drug and alcohol use among college student-athletes*. *Social Science Quarterly* 2015;96:369–80.
64 Dimeo P, Taylor J. Monitoring drug use in sport: the contrast between official statistics and other evidence. *Drugs: Education, Prevention and Policy* 2013;20:40–7.
65 Harcourt PR, Unglik H, Cook JL. A strategy to reduce illicit drug use is effective in elite Australian football. *Br J Sports Med* 2012;46:943–5.
66 Gaillard Y, Vayssette F, Pépin G. Compared interest between hair analysis and urinalysis in doping controls. Results for amphetamines, corticosteroids and anabolic steroids in racing cyclists. *Forensic Sci Int* 2000;107:1–3.
67 Simon P, Neuberger EWI, Wang G, *et al*. Antidoping science: important lessons from the medical sciences. *Curr Sports Med Rep* 2018;17:326–31.
68 Alquraini H, Auchus RJ. Strategies that athletes use to avoid detection of androgenic-anabolic steroid doping and sanctions. *Mol Cell Endocrinol* 2018;464:28–33.
69 LaBrie JW, Grossbard JR, Hummer JF. Normative misperceptions and marijuana use among male and female college athletes. *J Appl Sport Psychol* 2009;21:S77–S85.

Br J Sports Med: first published as 10.1136/bjsports-2019-100669 on 16 May 2019. Downloaded from http://bjsm.bmj.com/ on 19 May 2019 by guest. Protected by copyright.

70 Bonn-Miller MO, Loflin MJE, Thomas BF, *et al*. Labeling accuracy of cannabidiol extracts sold online. *JAMA* 2017;318:1708–9.
71 Möller I, Wintermeyer A, Bender K, *et al*. Screening for the synthetic cannabinoid JWH-018 and its major metabolites in human doping controls. *Drug Test Anal* 2011;3:609–20.
72 Heltsley R, Shelby MK, Crouch DJ, *et al*. Prevalence of synthetic cannabinoids in U.S. athletes: initial findings. *J Anal Toxicol* 2012;36:588–93.
73 Castaneto MS, Gorelick DA, Desrosiers NA, *et al*. Synthetic cannabinoids: epidemiology, pharmacodynamics, and clinical implications. *Drug Alcohol Depend* 2014;144:12–41.
74 Egan KL, Erausquin JT, Milroy JJ, *et al*. Synthetic cannabinoid use and descriptive norms among collegiate Student-Athletes. *J Psychoactive Drugs* 2016;48:166–72.
75 Strano Rossi S, Botrè F. Prevalence of illicit drug use among the Italian athlete population with special attention on drugs of abuse: a 10-year review. *J Sports Sci* 2011;29:471–6.
76 Cohen PA, Travis JC, Venhuis BJ. A methamphetamine analog (*N*,α -diethylphenylethylamine) identified in a mainstream dietary supplement. *Drug Test Anal* 2014;6:805–7.
77 Stogner JM, Miller BL. Investigating the 'bath salt' panic: The rarity of synthetic cathinone use among students in the United States. *Drug Alcohol Rev* 2013;79:n/a–49.
78 Conrad A, Hutton S, Munnelly M, *et al*. Screening for smokeless tobacco use and presence of oral lesions in major league baseball athletes. *J Calif Dent Assoc* 2015;43:14–20.
79 Alaranta A, Alaranta H, Patja K, *et al*. Snuff use and smoking in Finnish Olympic athletes. *Int J Sports Med* 2006;27:581–6.
80 Gouttebarge V, Aoki H, Kerkhoffs G. Prevalence and determinants of symptoms related to mental disorders in retired male professional footballers. *J Sports Med Phys Fitness* 2016;56:648–54.
81 Bents RT, Tokish JM, Ephedrine GL. Pseudoephedrine, and amphetamine prevalence in college hockey players: most report performance-enhancing use. *The Physician and Sportsmedicine* 2004;32:30–4.
82 Rexroat M. *NCAA national study of substance use habits of college student-athletes. alcohol abuse and drug use in sports and performance. Oxford research encyclopedia of psychology*. Oxford, UK: Oxford University Press, 2017.
83 Kersey RD, Elliot DL, Goldberg L, *et al*. National Athletic Trainers' Association position statement: anabolic-androgenic steroids. *J Athl Train* 2012;47:567–88.
84 Tricker R, O'Neill MR, Cook D. The incidence of anabolic steroid use among competitive bodybuilders. *J Drug Educ* 1989;19:313–25.
85 Wagman DF, Curry LA, Cook DL. An investigation into anabolic androgenic steroid use by elite US powerlifters. *Journal of Strength & Conditioning Research* 1995;9:149–54.
86 Horn S, Gregory P, Guskiewicz KM. Self-reported anabolic-androgenic steroids use and musculoskeletal injuries: findings from the center for the study of retired athletes health survey of retired NFL players. *Am J Phys Med Rehabil* 2009;88:192–200.
87 Sobal J, Marquart LF. Vitamin/mineral supplement use among athletes: a review of the literature. *Int J Sports Nutr* 1994;4:320–34.
88 Maughan RJ, Depiesse F, Geyer H, *et al*. The use of dietary supplements by athletes. *J Sports Sci* 2007;25:S103–13.
89 Baylis A, Cameron-Smith D, Burke LM. Inadvertent doping through supplement use by athletes: assessment and management of the risk in Australia. *Int J Sport Nutr Exerc Metab* 2001;11:365–83.
90 Heikkinen A, Alaranta A, Helenius I, *et al*. Use of dietary supplements in Olympic athletes is decreasing: a follow-up study between 2002 and 2009. *J Int Soc Sports Nutr* 2011;8:1–8.
91 Kim Y, Sohn D, Choi SM. Cultural difference in motivations for using social network sites: a comparative study of American and Korean college students. *Computers in Human Behavior* 2011;27:365–72.
92 Lun V, Erdman KA, Fung TS, *et al*. Dietary supplementation practices in Canadian high-performance athletes. *Int J Sport Nutr Exerc Metab* 2012;22:31–7.
93 McDuff DR. *Sports psychiatry: strategies for life balance and peak performance*. Washington, DC: American Psychiatric Publishing, 2012.
94 Walters P, Hearn A, Currie C. Substance misuse. In: Currie A, Owen B, eds. *Sports psychiatry. Oxford*. UK: Oxford University Press, 2016: 69–85.
95 Gouttebarge V, Frings-Dresen MHW, Sluiter JK. Mental and psychosocial health among current and former professional footballers. *Occup Med* 2015;65:190–6.
96 Gouttebarge V, Hopley P, Kerkhoffs G, *et al*. Symptoms of common mental disorders in professional rugby: an international observational descriptive study. *Int J Sports Med* 2017;38:864–70.
97 McDuff DR, Garvin M. Working with sports organizations and teams. *Int Rev Psychiatry* 2016;28:595–605.
98 Blank C, Kopp M, Niedermeier M, *et al*. Predictors of doping intentions, susceptibility, and behaviour of elite athletes: a meta-analytic review. *SpringerPlus* 2016;5:1–14.
99 Weaver CC, Martens MP, Cadigan JM, *et al*. Sport-related achievement motivation and alcohol outcomes: an athlete-specific risk factor among intercollegiate athletes. *Addict Behav* 2013;38:2930–6.
100 Zhou J, Heim D, Levy A. Sports participation and alcohol use: associations with sports-related identities and well-being. *J Stud Alcohol Drugs* 2016;77:170–9.
101 Storch EA, Kolsky AR, Silvestri SM, *et al*. Religiosity of elite college athletes. *The Sport Psychologist* 2001;15:346–51.
102 Veliz PT, Boyd CJ, McCabe SE. Competitive sport involvement and substance use among adolescents: a nationwide study. *Subst Use Misuse* 2015;50:156–65.
103 Barnes MJ. Alcohol: impact on sports performance and recovery in male athletes. *Sports Med* 2014;44:909–19.
104 Pesta DH, Angadi SS, Burtscher M, *et al*. The effects of caffeine, nicotine, ethanol, and tetrahydrocannabinol on exercise performance. *Nutr Metab* 2013;10.
105 Ware MA, Jensen D, Barrette A, *et al*. Cannabis and the health and performance of the elite athlete. *Clin J Sport Med* 2018;28:480–4.
106 Reardon CL, Factor RM. Considerations in the use of stimulants in sport. *Sports Med* 2016;46:611–7.
107 Kaufman KR. Modafinil in sports: ethical considerations. *Br J Sports Med* 2005;39:241–4.
108 Trinh KV, Kim J, Ritsma A. Effect of pseudoephedrine in sport: a systematic review. *BMJ Open Sport Exerc Med* 2015;1:e000066.
109 Cohen J, Collins R, Darkes J, *et al*. A league of their own: demographics, motivations and patterns of use of 1,955 male adult non-medical anabolic steroid users in the United States. *J Int Soc Sports Nutr* 2007;4:12–14.
110 Kanayama G, Kean J, Hudson JI, *et al*. Cognitive deficits in long-term anabolic-androgenic steroid users. *Drug Alcohol Depend* 2013;130:208–14.
111 Creado S, Reardon C. The sports psychiatrist and performance-enhancing drugs. *Int Rev Psychiatry* 2016;28:564–71.
112 Huang G, Basaria S. Do anabolic-androgenic steroids have performance-enhancing effects in female athletes? *Mol Cell Endocrinol* 2018;464:56–64.
113 Gwizdek K, Brzek A, Bak-Sosnowska M, *et al*. The use of steroids by gym athletes: an attempt to diagnose the problem scale and possible causes. *J Sports Med Phys Fitness* 2018;58:880–8.
114 MazzeoF, Mazzeo F, Mazzeo F. Anabolic steroid use in sports and in physical activity: overview and analysis. *Sport Mont* 2018;16:113–8.
115 Patel DR, Omar H, Terry M. Sport-related performance anxiety in young female athletes. *J Pediatr Adolesc Gynecol* 2010;23:325–35.
116 Paul MA, Gray G, Kenny G, *et al*. Impact of melatonin, zaleplon, zopiclone, and temazepam on psychomotor performance. *Aviat Space Environ Med* 2003;74:1263–70.
117 Charles RB, Kirkham AJ, Guyatt AR, *et al*. Psychomotor, pulmonary and exercise responses to sleep medication. *Br J Clin Pharmacol* 1987;24:191–7.
118 Johnston A, McAllister-Williams RH. Psychotropic drug prescribing. In: Currie A, Owen B, eds. *Sports psychiatry*. Oxford: Oxford University Press, 2016: 133–43.
119 Choi PYL, Pope HG, Olivardia R. Muscle dysmorphia: a new syndrome in weightlifters * commentary. *Br J Sports Med* 2002;36:375–6.
120 Esco MR, Olson MS, Williford HN. Muscle dysmorphia: an emerging body image concern in men. *Strength and Conditioning Journal* 2005;27:76–9.
121 Leone JE, Sedory EJ, Gray KA. Recognition and treatment of muscle dysmorphia and related body image disorders. *J Athl Train* 2005;40:352–9.
122 Walsh MM, Hilton JF, Masouredis CM, *et al*. Smokeless tobacco cessation intervention for college athletes: results after 1 year. *Am J Public Health* 1999;89:228–34.
123 McDuff DR, Morse ED, White RK. Professional and collegiate team assistance programs: services and utilization patterns. *Clin Sports Med* 2005;24:943–58.
124 Moesch K, Kenttä G, Kleinert J, *et al*. FEPSAC position statement: mental health disorders in elite athletes and models of service provision. *Psychology of Sport and Exercise* 2018;38:61–71.
125 Martens MP. Alcohol abuse and drug use in sport and performance. *Oxford Research Encyclopedia of Psychology*, 2017. Available: http://psychology.oxfordre.com/abstract/10.193/acrefore/9780190236557.001.0001/acrefore978019236557-e-168
126 Donohue B, Gavrilova Y, Galante M, *et al*. Empirical development of a screening method for mental, social, and physical wellness in amateur and professional Circus artists. *Psychology of Aesthetics, Creativity, and the Arts* 2018.
127 Donohue B, Loughran T, Pitts M, *et al*. Preliminary development of a brief intervention to prevent alcohol misuse and enhance sport performance in collegiate athletes. *J Drug Abuse* 2016;2.
128 Cimini MD, Monserrat JM, Sokolowski KL, *et al*. Reducing high-risk drinking among student-athletes: the effects of a targeted athlete-specific brief intervention. *J Am Coll Health* 2015;63:343–52.
129 Gross M, Moore ZE, Gardner FL, *et al*. An empirical examination comparing the Mindfulness-Acceptance-Commitment approach and psychological skills training for the mental health and sport performance of female student athletes. *Int J Sport Exerc Psychol* 2018;16:431–51.
130 Agley J, Walker BB, Gassman RA. Adaptation of alcohol and drug screening, brief intervention and referral to treatment (SBIRT) to a department of intercollegiate athletics: the COMPASS project. *Health Educ J* 2013;72:647–59.
131 Weiss RD, Jaffee WB, de Menil VP, *et al*. Group therapy for substance use disorders: what do we know? *Harv Rev Psychiatry* 2004;12:339–50.

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132 Gil F, de Andrade AG, Castaldelli-Maia JM, *et al*. Impacts, and treatment of substance use disorders in athletes. *Int Rev Psychiatry* 2016;28:572–8.
133 Woody GE. Advances in the treatment of opioid use disorders. *F1000Research* 2017;6.
134 Canadian Agency for Drugs and Technologies in Health (CADTH). Nicotine replacement therapy, bupropion and varenicline for tobacco cessation: a review of clinical effectiveness: CADTH, 2016. Available: https://www.cadth.ca/nicotine-replacement-therapy-bupropion-and-varenicline-tobacco-cessation-review-clinical [Accessed updated March 8, 2016.].
135 Ebbert JO, Hatsukami DK, Croghan IT, *et al*. Combination varenicline and bupropion SR for tobacco-dependence treatment in cigarette smokers. *JAMA* 2014;311:155–63.
136 Graznick N, Anthenelli RM. E-cigarettes and vapes: do they work for smoking cessation and should we be recommending their use? *Current Psychiatry* 2017;16:30-34–6.
137 Brose LS, Brown J, Hitchman SC, *et al*. Perceived relative harm of electronic cigarettes over time and impact on subsequent use. A survey with 1-year and 2-year follow-ups. *Drug Alcohol Depend* 2015;157:106–11.

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